

EXACT SPECTRA OF GENERALIZED CUBIC SUBDIVISION MATRICES

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ABSTRACT. The spectrum of a subdivision matrix controls local convergence near an extraordinary mesh element. For every surface valence $n \geq 3$, Dietz’s generalized cubic B-spline construction produces an initial-element matrix

$$S_n = \exp((\log 2)(\tilde{L}_n - I)) \exp((\log 2)(\tilde{R}_n - I)),$$

and a released double-ring matrix $\hat{S}_n$ assembled from S_n and regular cubic stencils. The source established two independent $1/2$ -eigenvectors but left open whether $1/2$ is subdominant, has exactly the expected multiplicity, and is semisimple. We resolve all three questions and give the complete spectra of both released surface matrices. A discrete Fourier transform reduces S_n to one 3×3 block and $n - 1$ blocks of size 2×2 ; a phase-free block-triangular form gives the outer spectrum

$$\{0^{[24n]}, (1/64)^{[n]}, (1/32)^{[n]}, (1/16)^{[n]}, (1/8)^{[n]}\},$$

where brackets denote algebraic multiplicity. Thus both matrices have a simple dominant eigenvalue 1, a semisimple subdominant eigenvalue $1/2$ of algebraic and geometric multiplicity two, and no other eigenvalue of modulus at least $1/2$. For $n \geq 5$, the subsubdominant eigenvalue satisfies

$$\frac{1}{2} - \frac{3\pi^2}{4n^2} + \frac{\pi^4}{2n^4} + O(n^{-6})$$

as $n \rightarrow \infty$, explaining the vanishing spectral gap. A proved tensor identity gives the expected multiplicity three for the semi-regular initial-volume family. The analogous double-shell conclusion remains conditional on the tensor identity stated with a proof sketch in Dietz’s source. Exact-identity checks, independent reconstructions, and Lean certificates accompany the paper. Fully irregular three-dimensional matrices and a complete C^1 theorem are not covered.

1. INTRODUCTION

The local behavior of a stationary subdivision scheme near an extraordinary element is governed by the spectrum of its subdivision matrix. In the usual smoothness analysis, one seeks a simple dominant eigenvalue 1, a subdominant eigenvalue with the dimensionally expected multiplicity, and

Date: August 14, 2026.

2020 Mathematics Subject Classification. 65D17, 15A18, 41A15.

Key words and phrases. subdivision surfaces, generalized cubic B-splines, extraordinary vertices, Fourier analysis, matrix exponentials, exact spectrum.

a characteristic map built from the corresponding eigenspace; see, for example, [3, 8, 1, 15, 12, 13, 14]. Fourier block reduction, exact eigen-expansion, and eigencomponent tuning are standard tools in this setting [16, 9, 2]. For a surface scheme the expected multiplicity is two, while for a volumetric scheme it is three.

Dietz introduced generalized quadratic and cubic B-spline subdivision matrices for surfaces and volumes in the authoritative German dissertation [4]; for accessibility, most section references below also cite its author-released English machine translation [5]. In the cubic construction the final matrix is a product of two matrix exponentials. Dietz proved that the coordinate columns are $1/2$ -eigenvectors, but explicitly left open whether $1/2$ is subdominant with algebraic multiplicity two or three [4, Thms. 6.30–6.32, pp. 251–252]. The dissertation also identifies cyclic Fourier analysis as the natural route for the surface case [5, Sec. 6.3]. The released implementation contains both the $(2n + 1)$ -dimensional initial-element matrix and a $(30n + 1)$ -dimensional double-ring matrix assembled from the central rule and regular cubic stencils [6]. The broader approximation-theory program treats subdivision as a route to spline spaces on irregular meshes and highlights unresolved analysis questions [7, 11]. Recent averaging-and-refinement constructions can also prescribe a half-eigenvalue, but define different schemes and matrices [18].

This note carries out that analysis. Beyond the qualitative assertion left unresolved in the cited source, it gives every eigenvalue of both released surface matrices explicitly. The proof has five ingredients.

- (1) Direct evaluation of Dietz’s local Colin-de-Verdière-like square mask [5, Sec. 6.2.2] gives a three-layer block-circulant matrix.
- (2) Cyclic symmetry decomposes its complexification into Fourier modes.
- (3) Every nonzero mode has a quadratic characteristic polynomial whose two tangent modes factor at $1/2$, while every other nonzero mode obeys a strict unit-disk criterion after scaling by two.
- (4) The constant mode has a cubic characteristic polynomial that factors exactly into three linear factors.
- (5) A topological permutation puts the outer part of the double-ring matrix in block upper-triangular form, with copies of one fixed 3×3 block and one scalar block on the diagonal and a strictly upper-triangular zero-spectral block.

The fully irregular volumetric family is not block-circulant. We therefore prove the claimed spectral properties for both released surface categories and the proved semi-regular initial-volume tensor product, while recording the double-shell consequence conditionally and leaving the genuinely irregular volume problem separate. This scope distinction is important: the argument below resolves the cited surface spectral question, not an arbitrary three-dimensional polyhedral-input problem.

The spectral theorem removes one algebraic obstacle in the standard subdivision-smoothness pipeline: the tangent eigenspace now has exactly the required dimension, and the remaining spectrum is strictly separated. The gap is nevertheless only of order n^{-2} , so high valence produces slow spectral separation. Characteristic-map regularity and injectivity, and hence a complete C^1 theorem, still require separate geometric arguments; the present result supplies none implicitly [15, 13].

The all-valence conclusions come from the displayed symbolic block factorizations and inequalities. Finite dense eigensystem sweeps are used only as implementation QA. Source hashes identify the matrices being analyzed; they do not substitute for the source-to-matrix derivation given in the paper.

2. THE CUBIC SURFACE MATRIX

Fix an integer $n \geq 3$, set

$$\theta = \frac{\pi}{n}, \quad s = \sin \theta, \quad c = \cos \theta,$$

and index all cyclic quantities modulo n . Let

$$e_j = (\cos 2j\theta, \sin 2j\theta), \quad p_j = \frac{1}{c}(\cos(2j+1)\theta, \sin(2j+1)\theta), \quad f = 0.$$

Thus e_j is the tangent point of the unit circle on the side between the adjacent corners p_{j-1} and p_j . The control vector is ordered as

$$(P, E, f) = (p_0, \dots, p_{n-1}, e_0, \dots, e_{n-1}, f).$$

Let $Z \in \mathbb{R}^{n \times n}$ be the cyclic shift

$$(Zx)_j = x_{j+1}.$$

We now make the local weight rule explicit. For shifted cell vertices $w_0, \dots, w_3 \in \mathbb{R}^2$ in cyclic order, Dietz's Colin-de-Verdière-like construction assigns the raw off-diagonal entry

$$(1) \quad d_{i\ell} = -\frac{1}{|\det(w_i, w_\ell)|} \quad \text{if } \ell = i \pm 1, \quad d_{i\ell} = 0 \quad \text{otherwise.}$$

The diagonal d_{ii} is chosen so that $\sum_\ell d_{i\ell} w_\ell = 0$, and the row is then divided by $\sum_\ell d_{i\ell}$ so that it sums to one [5, Sec. 6.2.2]; the same rule is implemented in the released routine [6]. For the square below, the two coordinate equations determining the diagonal are compatible; the following calculation verifies compatibility, nonzero row sums, and the resulting normalized mask directly.

Lemma 2.1 (Local square mask). *For the square (p_j, e_j, f, e_{j+1}) , shifted by $p_j/2$, the normalized local matrix in Dietz's construction is*

$$M_{\square} = \begin{pmatrix} 1 - \frac{1}{2s^2} & \frac{1}{4s^2} & 0 & \frac{1}{4s^2} \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & \frac{1}{4c^2} & 1 - \frac{1}{2c^2} & \frac{1}{4c^2} \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 \end{pmatrix}.$$

Every row sums to one, and

$$M_{\square} \begin{pmatrix} p_j - p_j/2 \\ e_j - p_j/2 \\ f - p_j/2 \\ e_{j+1} - p_j/2 \end{pmatrix} = 0.$$

Proof. Rotate the configuration so that $p_j = (1/c, 0)$, $e_j = (c, -s)$, and $e_{j+1} = (c, s)$. After the shift by $p_j/2$, write

$$\begin{aligned} w_0 &= \left(\frac{1}{2c}, 0 \right), & w_1 &= \left(\frac{\cos 2\theta}{2c}, -s \right), \\ w_2 &= \left(-\frac{1}{2c}, 0 \right), & w_3 &= \left(\frac{\cos 2\theta}{2c}, s \right). \end{aligned}$$

Every adjacent determinant $|\det(w_i, w_{i+1})|$ equals $s/(2c)$, so (1) gives the raw off-diagonal edge entry $m = -2c/s$. For w_1, w_3 the two neighbours cancel because $w_0 + w_2 = 0$, so both coordinate equations give $d_1 = d_3 = 0$. For w_0, w_2 the y -equation is identically zero, since w_1, w_3 have opposite y -coordinates; the x -equation alone gives the compatible raw diagonals

$$d_0 = \frac{4c \cos 2\theta}{s}, \quad d_1 = d_3 = 0, \quad d_2 = -\frac{4c \cos 2\theta}{s}.$$

The four raw row sums are

$$-8cs, \quad -\frac{4c}{s}, \quad -\frac{8c^3}{s}, \quad -\frac{4c}{s},$$

where $1 - \cos 2\theta = 2s^2$ and $1 + \cos 2\theta = 2c^2$ were used. These sums are nonzero also at $n = 4$, where $\cos 2\theta = 0$. Dividing each row by its sum gives exactly the displayed mask. Row stochasticity is then automatic, and multiplication by the shifted coordinate columns reduces to $s^2 + c^2 = 1$. $\square$

The corner copy occurs once, an edge copy twice, and the center copy n times. Averaging the copies gives the global matrix.

Lemma 2.2 (Assembled initial matrix). *The matrix before spectral adjustment is*

$$(2) \quad C_n = \begin{pmatrix} \left(1 - \frac{1}{2s^2}\right) I & \frac{1}{4s^2}(I + Z) & 0 \\ \frac{1}{4}(I + Z^*) & 0 & \frac{1}{2}\mathbf{1} \\ 0 & \frac{1}{2nc^2}\mathbf{1}^* & 1 - \frac{1}{2c^2} \end{pmatrix}.$$

In particular, $C_n\mathbf{1} = \mathbf{1}$ and

$$C_n(P, E, f)^\top = \frac{1}{2}(P, E, f)^\top.$$

Proof. The corner row is the first row of the local mask. Each edge row is the average of the two second-or-fourth local rows containing that edge, and the center row is the average of the n center rows. This gives (2). The reproduction identities also follow directly from Lemma 2.1 and the averaging. $\square$

Dietz doubles C_n , puts its strict lower and strict upper parts into separate matrices, and chooses the two diagonals so that both matrices have row sums equal to one. With the layer ordering above,

$$(3) \quad \begin{aligned} \tilde{L}_n &= \begin{pmatrix} I & 0 & 0 \\ \frac{1}{2}(I + Z^*) & 0 & 0 \\ 0 & \frac{1}{nc^2}\mathbf{1}^* & -\tan^2 \theta \end{pmatrix}, \\ \tilde{R}_n &= \begin{pmatrix} -\cot^2 \theta I & \frac{1}{2s^2}(I + Z) & 0 \\ 0 & 0 & \mathbf{1} \\ 0 & 0 & 1 \end{pmatrix}. \end{aligned}$$

Their off-diagonal entries partition those of $2C_n$; since $C_n\mathbf{1} = \mathbf{1}$, the row-sum-one diagonal conditions force their diagonal entries to sum to the diagonal of $2C_n$. Hence their sum is $2C_n$. The subdivision matrix is

$$(4) \quad S_n = L_n R_n, \quad L_n = \exp((\log 2)(\tilde{L}_n - I)), \quad R_n = \exp((\log 2)(\tilde{R}_n - I)).$$

Equations (2)–(4) are also a closed-form re-expression of the released MATLAB surface routine.

Source fidelity. The relation between the mathematical matrices defined here and the released implementation is an audited provenance statement, not part of the spectral theorems. The reconstruction uses the frozen repository tree

6be55ee3e88195619415eb9f6ea1901324e239fc.

The decisive files, authenticated in full by both Git-blob SHA-1 and SHA-256, are:

routine	Git blob	SHA-256 prefix
<code>computeBiCubicSubdivisionMatrix.m</code>	<code>bb00f5f34260</code>	<code>f64047ed9df1480d</code>
<code>computeBiCubicSubdivisionMatrixBig.m</code>	<code>ecf810899065</code>	<code>82767dc59f57bdf1</code>
<code>computeTriCubicSubdivisionMatrix.m</code>	<code>350691ce24c3</code>	<code>5da5f541ef3f6e2e</code>

The complete digests and exact source fragments occur in the machine-readable source manifest. For the initial routine, an independent implementation reconstructs the matrix both by local-cell assembly and from (2)–(4); the routes agree entrywise. For the double-ring routine, the complete-file authentication covers the central insertion and every generic and wrap-around assignment used in the support decomposition below. A separately organized implementation translates all released rows from the four displayed rational stencils and checks both the central block and the spectral union. Thus the hashes identify the source version, while the displayed derivations and independent reconstructions establish the mathematical correspondence.

3. FOURIER REDUCTION

Let $\zeta = e^{2\pi i/n}$, and for $0 \leq k < n$ define

$$\phi_k = (1, \zeta^k, \zeta^{2k}, \dots, \zeta^{(n-1)k})^\top.$$

Then $Z\phi_k = \zeta^k \phi_k$. For $k \neq 0$, the two-dimensional subspace

$$\mathcal{H}_k = \text{span}\{(\phi_k, 0, 0), (0, \phi_k, 0)\}$$

is invariant under $\tilde{L}_n$, $\tilde{R}_n$, and S_n . The constant subspace

$$\mathcal{H}_0 = \text{span}\{(\phi_0, 0, 0), (0, \phi_0, 0), (0, 0, 1)\}$$

has dimension three. Hence

$$(5) \quad \mathbb{C}^{2n+1} = \mathcal{H}_0 \oplus \mathcal{H}_1 \oplus \dots \oplus \mathcal{H}_{n-1}.$$

For a nonzero mode define

$$(6) \quad \alpha_k = \frac{1 + \zeta^{-k}}{2}, \quad \beta_k = \frac{1 + \zeta^k}{2s^2}, \quad \rho_k = \alpha_k \beta_k = \frac{\cos^2(k\theta)}{s^2}.$$

Also put

$$(7) \quad u = \csc^2 \theta, \quad x = 2^{-u}, \quad g = \frac{\frac{1}{2} - x}{u - 1}.$$

Since $u > 1$, one has $0 < x < 1/2$ and $g > 0$.

Lemma 3.1 (Nonzero Fourier blocks). *On $\mathcal{H}_k$, $k \neq 0$, the two exponential factors and their product are*

$$(8) \quad L_k = \begin{pmatrix} 1 & 0 \\ \alpha_k/2 & 1/2 \end{pmatrix}, \quad R_k = \begin{pmatrix} x & \beta_k g \\ 0 & 1/2 \end{pmatrix},$$

$$(9) \quad S_k = L_k R_k = \begin{pmatrix} x & \beta_k g \\ \alpha_k x/2 & 1/4 + \rho_k g/2 \end{pmatrix}.$$

Its characteristic polynomial is

$$(10) \quad \chi_k(\lambda) = \lambda^2 - T_k \lambda + \frac{x}{4}, \quad T_k = x + \frac{1}{4} + \frac{\rho_k g}{2}.$$

Proof. On $\mathcal{H}_k$, the generators in (4) are

$$\begin{pmatrix} 0 & 0 \\ \alpha_k & -1 \end{pmatrix}, \quad \begin{pmatrix} -u & \beta_k \\ 0 & -1 \end{pmatrix}.$$

For either triangular block $T = D + N$, direct solution of $X'(t) = (T - I)X(t)$ with $X(0) = I$ gives the diagonal terms $e^{t(d_i-1)}$ and the off-diagonal divided difference $N_{ij}(e^{t(d_i-1)} - e^{t(d_j-1)})/(d_i - d_j)$ (with its continuous equal-diagonal limit). Substitution of $t = \log 2$ gives (8); multiplication and a 2×2 determinant give (9) and (10). $\square$

The discriminant is always nonnegative:

$$(11) \quad T_k^2 - x \geq \left(x + \frac{1}{4}\right)^2 - x = \left(x - \frac{1}{4}\right)^2.$$

Moreover, $T_k > 0$ and the two roots have product $x/4 > 0$. A real pair with positive product and positive sum T_k consists of two positive roots. Thus every nonzero-mode eigenvalue is real and positive.

4. THE TANGENT MODES AND STRICT SUBDOMINANCE

Lemma 4.1 (The two half-eigenvalues). *For $k = 1$ and $k = n - 1$,*

$$(12) \quad \chi_k(\lambda) = \left(\lambda - \frac{1}{2}\right) \left(\lambda - \frac{x}{2}\right).$$

In each of these two modes, $1/2$ is a simple eigenvalue.

Proof. For $k = 1, n - 1$, (6) gives $\rho_k = \cot^2 \theta = u - 1$. Hence

$$\frac{\rho_k g}{2} = \frac{1}{4} - \frac{x}{2}, \quad T_k = \frac{1}{2} + \frac{x}{2},$$

and (12) follows. Since $x < 1/2$, its two roots are distinct. $\square$

Lemma 4.2 (All other nonzero modes lie below one half). *If $k \notin \{0, 1, n-1\}$, then every root of χ_k lies in $(0, 1/2)$.*

Proof. For these modes,

$$(13) \quad \rho_k < \cot^2 \theta,$$

For $n = 3$ the index range is empty. For $n \geq 4$ and $2 \leq k \leq n - 2$, $-\cos \theta < \cos(k\theta) < \cos \theta$, by symmetry about $\pi/2$ and strict monotonicity of cosine on $[0, \pi]$; hence $|\cos(k\theta)| < \cos \theta$. Using (7),

$$(14) \quad 1 - 2T_k + x = \left(\frac{1}{2} - x\right) \left(1 - \frac{\rho_k}{\cot^2 \theta}\right) > 0.$$

Consequently,

$$\chi_k(1/2) = \frac{1}{4}(1 - 2T_k + x) > 0.$$

The two roots are real and positive by (11), and their product is $x/4 < 1/8$. Since the upward-opening polynomial is positive at $1/2$, either both roots are below $1/2$ or both are above it. The latter is impossible because their product would then exceed $1/4$. $\square$

Equivalently, after scaling $\mu = 2\lambda$, the quadratic is

$$\mu^2 - 2T_k\mu + x,$$

and (14), together with $1 + 2T_k + x > 0$ and $1 - x > 0$, is the strict second-order Jury criterion.

5. THE CONSTANT FOURIER MODE

Set

$$v = \sec^2 \theta, \quad q = e^{-t}, \quad x_t = e^{-ut}, \quad y_t = e^{-vt}.$$

The identity $s^2 + c^2 = 1$ is equivalent to

$$(15) \quad \frac{1}{u} + \frac{1}{v} = 1, \quad v = \frac{u}{u-1}.$$

On $\mathcal{H}_0$, the two generators are

$$(16) \quad A_0 = \begin{pmatrix} 0 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & v & -v \end{pmatrix}, \quad B_0 = \begin{pmatrix} -u & u & 0 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

For arbitrary $t > 0$, their exponentials are

$$(17) \quad E_L(t) = \begin{pmatrix} 1 & 0 & 0 \\ 1-q & q & 0 \\ 1-uq+(u-1)y_t & u(q-y_t) & y_t \end{pmatrix},$$

$$(18) \quad E_R(t) = \begin{pmatrix} x_t & v(q-x_t) & 1-x_t-v(q-x_t) \\ 0 & q & 1-q \\ 0 & 0 & 1 \end{pmatrix}.$$

Lemma 5.1 (Constant-mode factorization). *For every $t > 0$,*

$$(19) \quad \det(\lambda I - E_L(t)E_R(t)) = (\lambda - 1)(\lambda - q^2)(\lambda - x_t y_t).$$

At $t = \log 2$, the constant-mode eigenvalues are therefore

$$(20) \quad 1, \quad \frac{1}{4}, \quad 2^{-(\csc^2 \theta + \sec^2 \theta)}.$$

Proof. Equations (17) and (18) follow by solving the two triangular linear systems, or by direct exponentiation. Both factors are row-stochastic, so $E_L(t)E_R(t)\mathbf{1} = \mathbf{1}$. A second eigenvector is

$$w_q = \left(-\frac{1}{q}, \frac{q-u+1}{qu}, 1 \right)^\top,$$

and direct multiplication using (15) gives $E_L(t)E_R(t)w_q = q^2 w_q$. Finally,

$$\det(E_L(t)E_R(t)) = (q y_t)(x_t q) = q^2 x_t y_t.$$

The characteristic polynomial is monic of degree three. Since $t > 0$ gives $q = e^{-t} \in (0, 1)$ and hence $q^2 \neq 1$, after removing the known factors $\lambda - 1$ and $\lambda - q^2$ its remaining root is $x_t y_t$. This proves (19). The ancillary symbolic certificate and the Lean development independently expand and check the same identity. $\square$

Since

$$(21) \quad \csc^2 \theta + \sec^2 \theta = \frac{1}{s^2 c^2} \geq 4,$$

one has $x_t y_t = q^{u+v} \leq q^4 < q^2 < 1$ for every $t > 0$. At $t = \log 2$, the last eigenvalue in (20) is at most $1/16$. Hence the dominant eigenvalue 1 is simple, and the constant mode contributes no eigenvalue of modulus $1/2$.

6. COMPLETE INITIAL-ELEMENT SPECTRUM

Theorem 6.1 (Exact subdominant half-spectrum). *For every integer $n \geq 3$, the generalized cubic surface subdivision matrix S_n defined by (2)–(4) has the following properties.*

- (1) 1 is a simple eigenvalue.
- (2) $1/2$ is an eigenvalue of algebraic and geometric multiplicity exactly two.
- (3) $1/2$ is semisimple; no nontrivial Jordan block occurs at $1/2$.
- (4) Every other eigenvalue λ satisfies $0 < \lambda < 1/2$.
- (5) S_n is diagonalizable over $\mathbb{R}$.

More precisely, counted with algebraic multiplicity,

$$(22) \quad \text{spec}(S_n) = \left\{1, \frac{1}{4}, 2^{-(u+v)}\right\} \cup \bigcup_{k=1}^{n-1} \left\{ \frac{T_k + \sqrt{T_k^2 - x}}{2}, \frac{T_k - \sqrt{T_k^2 - x}}{2} \right\},$$

where

$$(23) \quad u = \csc^2 \theta, \quad v = \sec^2 \theta, \quad x = 2^{-u}, \quad T_k = x + \frac{1}{4} + \frac{\cos^2(k\theta)}{2s^2} \frac{\frac{1}{2} - x}{u - 1}.$$

Proof. The direct sum (5) accounts for all $2n + 1$ dimensions. Lemma 5.1 gives the three constant-mode eigenvalues. Lemma 4.1 gives exactly one $1/2$ -eigenvalue in each of the two tangent modes $k = 1, n - 1$, with the second eigenvalue $x/2 < 1/4$. Lemma 4.2 puts every other nonzero-mode eigenvalue strictly below $1/2$. This proves the spectral multiset, the algebraic multiplicities, and (22).

Every nonzero block has two distinct roots except possibly when equality holds throughout (11). That requires $\rho_k = 0$ and $x = 1/4$, hence $n = 4$ and $k = 2$. In this exceptional case $\alpha_k = \beta_k = 0$, and (9) is exactly $(1/4)I_2$, so it is still diagonalizable. The three constant-mode eigenvalues are distinct by (21). Therefore every Fourier block is diagonalizable. The full matrix is diagonalizable over $\mathbb{C}$; because it is real and all its eigenvalues are real, it is diagonalizable over $\mathbb{R}$. In particular, the two $1/2$ -eigenvalues are semisimple. $\square$

Corollary 6.2 (Completeness of the coordinate eigenspace). *The two coordinate columns $(P, E, f)^\top$ span the entire $1/2$ -eigenspace of S_n .*

Proof. Dietz's generator identities give

$$\tilde{L}_n(P, E, f)^\top = (P, E, f)^\top, \quad \tilde{R}_n(P, E, f)^\top = 0.$$

Thus the coordinate columns are independent $1/2$ -eigenvectors. [Theorem 6.1](#) says the eigenspace has dimension exactly two. $\square$

7. THE RELEASED DOUBLE-RING SURFACE MATRIX

The released double-ring routine constructs a matrix $\hat{S}_n$ of order $30n + 1$ for a double ring [6]. It inserts a permuted copy of S_n on the $2n + 1$ central indices and fills the remaining $28n$ rows with regular cubic tensor-product stencils. Every central row, indexed by $\{1\} \cup \{30_s, 31_s : s \in \mathbb{Z}/n\mathbb{Z}\}$, has zero entries in every outer column indexed by $\{2_s, \dots, 29_s\}$. This follows because the released assignment overwrites each complete central row with the corresponding row of S_n . Therefore a simultaneous row-column permutation gives

$$(24) \quad P_n \hat{S}_n P_n^{-1} = \begin{pmatrix} Q_n & B_n \\ 0 & S_n \end{pmatrix}.$$

The released routine uses the following four row stencils:

$$(25) \quad \begin{aligned} M_1 &= (1/4, 1/4, 1/4, 1/4), \\ M_2 &= (1/16, 3/8, 1/16, 1/16, 3/8, 1/16), \\ M_{22} &= (1/16, 1/16, 3/8, 3/8, 1/16, 1/16), \\ M_3 &= (1/64, 3/32, 1/64, 3/32, 9/16, 3/32, 1/64, 3/32, 1/64). \end{aligned}$$

Write r_s for local label $r \in \{2, \dots, 29\}$ in sector $s \in \mathbb{Z}/n\mathbb{Z}$. The labels $1, 30_s, 31_s$ are central and are deleted when forming Q_n . Figure 1 records the index layout used by the source. Labels 2–15 are outer tensor-grid rows; 16–22 and 26–28 are transition/connector rows; 23–25 and 29 form the nonzero outer core; and 30, 31 belong to the central copy of S_n . Sector seams identify the indicated neighbours with the corresponding labels in $s \pm 1$.

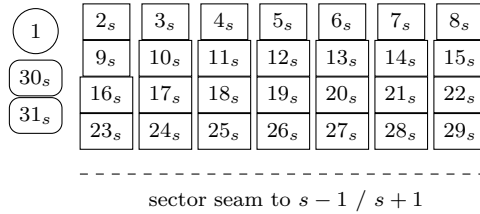


FIGURE 1. One-sector indexing schematic for the released double ring. The diagram fixes labels and sector gluing; the exact row supports and weights are stated in Lemma 7.1.

Lemma 7.1 (Outer support). *Translating the released column lists and then restricting to the outer columns gives the following support inclusions; “contained in” permits different rows in a grouped line to use different subsets.*

<i>outer row(s)</i>	<i>nonzero outer columns are contained in</i>
$2_s, \dots, 15_s$	$16_s, 17_s, 18_s, 19_s, 22_s, 23_s, 24_s, 25_s, 28_s, 29_s, 28_{s+1}, 29_{s+1}$
$16_s, \dots, 19_s$	$23_s, 24_s, 25_s, 29_s, 29_{s+1}$
$22_s, 28_s$	$23_s, 24_s, 29_s, 25_{s-1}$
$20_s, 21_s, 26_s, 27_s$	$19_{s-1}, 25_{s-1}, 22_s, 23_s, 24_s, 28_s, 29_s$
23_s	$23_s, 24_s, 25_s, 29_s, 25_{s-1}$
24_s	$24_s, 25_s$
25_s	$24_s, 25_s, 29_{s+1}$
29_s	$25_{s-1}, 29_s$

For the last four lines, the relevant outer weights, in the displayed column order, are respectively

$$\begin{aligned} r = 23 : & \quad (1/64, 3/32, 1/64, 3/32, 1/64), & r = 24 : & \quad (1/16, 1/16), \\ r = 25 : & \quad (1/64, 3/32, 1/64), & r = 29 : & \quad (1/16, 1/16). \end{aligned}$$

All omitted stencil entries multiply central columns. This table exposes the support information used below without requiring access to the MATLAB file. The spectral conclusion below depends only on these inclusions and the four exact weight rows for $r \in \{23, 24, 25, 29\}$.

Proof. Substituting the four stencils in (25) into each generic, wrap-around, and central-insertion row of the authenticated source gives the displayed column sets. Removing central columns leaves the stated inclusions; collecting the noncentral entries in rows 23, 24, 25, and 29 gives the four displayed rational weight lists. Thus harmless deletion of additional non-core entries preserves the dependency hierarchy used below. $\square$

Theorem 7.2 (Outer double-ring spectrum). *For every $n \geq 3$,*

$$\text{spec}(Q_n) = \left\{ 0^{[24n]}, (1/64)^{[n]}, (1/32)^{[n]}, (1/16)^{[n]}, (1/8)^{[n]} \right\},$$

where superscripts in square brackets denote algebraic multiplicity. Equivalently,

$$\chi_{Q_n}(\lambda) = \lambda^{24n}(\lambda - 1/64)^n(\lambda - 1/32)^n(\lambda - 1/16)^n(\lambda - 1/8)^n.$$

Proof. The support table shows that the three-vertex sets

$$\mathcal{C}_s = \{24_s, 25_s, 29_{s+1}\} \quad (s \in \mathbb{Z}/n\mathbb{Z})$$

are the only nontrivial strongly connected components. On each component, in the displayed order, the restricted matrix is the fixed real matrix

$$H = \begin{pmatrix} 1/16 & 1/16 & 0 \\ 1/64 & 3/32 & 1/64 \\ 0 & 1/16 & 1/16 \end{pmatrix}.$$

Each $\{23_s\}$ is a singleton component with diagonal entry $1/64$ and edges only toward $\mathcal{C}_s \cup \mathcal{C}_{s-1}$. Here an edge $r \rightarrow c$ means that row r has a nonzero entry in column c . The remaining $24n$ vertices have zero diagonal and form singleton components. Indeed, the table gives the acyclic dependency hierarchy

$$\begin{array}{lll} \mathcal{C}_s; & 23_s; & 16_s, \dots, 19_s, 22_s, 28_s; \\ & 20_s, 21_s, 26_s, 27_s; & 2_s, \dots, 15_s, \end{array}$$

where every non-core edge points to its own or an earlier displayed level (and no non-core row has a self-edge). A topological permutation therefore can be generated without Fourier analysis: list sectors in increasing order within each of the five groups

$$(2:15), \quad (20, 21, 26, 27), \quad (16:19, 22, 28), \quad (23), \quad (\mathcal{C}),$$

and concatenate the groups from left to right. Within the last group write each component as $(24_s, 25_s, 29_{s+1})$. This explicit ordering puts Q_n in block upper-triangular form

$$\Pi_n Q_n \Pi_n^{-1} = \begin{pmatrix} N_n & * & * \\ 0 & (1/64)I_n & * \\ 0 & 0 & H^{\oplus n} \end{pmatrix},$$

where N_n is strictly upper triangular of order $24n$. The stars record one-way intercomponent couplings; the claim is block triangularity, not block diagonality.

Direct expansion gives

$$\det(\lambda I - H) = \left(\lambda - \frac{1}{8}\right) \left(\lambda - \frac{1}{16}\right) \left(\lambda - \frac{1}{32}\right).$$

Together with $\chi_{N_n}(\lambda) = \lambda^{24n}$, the determinant of the block-triangular matrix is precisely the polynomial stated in the theorem. No Fourier phase is required for the outer block. $\square$

Corollary 7.3 (Double-ring half-spectrum). *For every $n \geq 3$,*

$$\text{spec}(\hat{S}_n) = \text{spec}(S_n) \uplus \text{spec}(Q_n).$$

Consequently, $\hat{S}_n$ has a simple dominant eigenvalue 1, a semisimple subdominant eigenvalue $1/2$ of algebraic and geometric multiplicity exactly two, and no other eigenvalue of modulus at least $1/2$.

Proof. The multiset identity follows from the block-triangular form (24). Theorem 7.2 gives $\rho(Q_n) = 1/8$, so neither 1 nor $1/2$ belongs to its spectrum. Hence the algebraic multiplicities at those values come entirely from S_n . Moreover, fix $\lambda \in \{1, 1/2\}$. Projection onto the lower coordinate block intertwines $P_n \hat{S}_n P_n^{-1}$ with S_n . On the λ -primary subspace this projection is injective: a vector in its kernel would lie in a generalized λ -eigenspace of Q_n , which is zero because $Q_n - \lambda I$ is invertible. The two primary subspaces have the same dimension, since the characteristic polynomial factorization gives the same algebraic multiplicity at λ ; hence the projection is an intertwining

isomorphism. Thus the Jordan structure at 1 and $1/2$ is exactly that of S_n . [Theorem 6.1](#) completes the proof. $\square$

8. THE CLOSING SPECTRAL GAP

Proposition 8.1 (Subsubdominant modes and asymptotic). *For $n \geq 5$, the largest eigenvalue outside $\{1, 1/2, 1/2\}$ occurs in the modes $k = 2, n-2$. For*

$$r_+(T) = \frac{T + \sqrt{T^2 - x}}{2}, \quad \frac{dr_+}{dT} = \frac{1}{2} \left(1 + \frac{T}{\sqrt{T^2 - x}} \right) > 0,$$

the larger root is strictly increasing in T , and T_k is strictly increasing in ρ_k . By the symmetry $\cos^2(k\theta) = \cos^2((n-k)\theta)$ and the strict decrease of cosine on $[0, \pi/2]$, the largest ρ_k outside $k = 1, n-1$ occurs precisely at $k = 2, n-2$. The constant-mode competitors are at most $1/4$, and the second tangent-mode root satisfies $x/2 < 1/4$ because $x < 1/2$, whereas

$$\chi_2(1/4) = -\frac{\rho_2 g}{8} < 0$$

for $n \geq 5$, so the larger mode-2 root exceeds $1/4$.

Let μ_n denote this subsubdominant eigenvalue. Since

$$(26) \quad \frac{\rho_2 g}{2} = \left(\frac{1}{4} - \frac{x}{2} \right) \frac{\cos^2(2\theta)}{\cos^2 \theta},$$

and $x = 2^{-\csc^2 \theta} = O(\theta^N)$ for every fixed N as $\theta \rightarrow 0$,

$$\frac{\cos^2(2\theta)}{\cos^2 \theta} = 1 - 3\theta^2 + 2\theta^4 + O(\theta^6).$$

Writing T_2 for the sum of the two mode-2 roots, their product $x/4$ gives the exact identity $\mu_n = T_2 - x/(4\mu_n)$. Since $\mu_n > 1/4$, this yields $|\mu_n - T_2| \leq x$ and hence $\mu_n = T_2 + O(2^{-1/\theta^2})$. Expanding $T_2 = \frac{1}{4}(1 + \Lambda) + x(1 - \Lambda/2)$ with $\Lambda = \cos^2(2\theta)/\cos^2 \theta$ gives

$$(27) \quad \mu_n = \frac{1}{2} - \frac{3}{4}\theta^2 + \frac{1}{2}\theta^4 + O(\theta^6) = \frac{1}{2} - \frac{3\pi^2}{4n^2} + \frac{\pi^4}{2n^4} + O(n^{-6}).$$

The subsubdominant value has algebraic multiplicity two, one copy in each of the modes $2, n-2$.

Thus the strict inequality in [Theorem 6.1](#) is sharp but not uniform in valence. At $n = 100$, the three-term expression predicts 0.4992602667, consistent with the empirical maximum 0.49926 reported for both surface categories in the German source [4, p. 252]. Since every outer double-ring eigenvalue has modulus at most $1/8$, [Corollary 7.3](#) also shows that S_n and $\hat{S}_n$ have exactly the same dominant, subdominant, and subsubdominant eigenvalues. This explains the equality of the relevant spectra reported for the initial and double-ring surface categories.

9. SEMI-REGULAR VOLUMETRIC MATRICES

The one-dimensional factor depends on whether the two-dimensional matrix describes an initial element or a double ring. This distinction is explicit in Dietz's tensor-product theorem [5, Thm. 2.58]. For the initial element, let

$$(28) \quad A = \frac{1}{8} \begin{pmatrix} 4 & 4 & 0 \\ 1 & 6 & 1 \\ 0 & 4 & 4 \end{pmatrix}.$$

Its characteristic polynomial is

$$(29) \quad \det(\lambda I - A) = (\lambda - 1) \left(\lambda - \frac{1}{2} \right) \left(\lambda - \frac{1}{4} \right).$$

For the double ring, the factor is instead the following eleven-state matrix:

$$(30) \quad B = \frac{1}{8} \begin{pmatrix} 0 & 0 & 4 & 4 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 6 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 & 4 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 6 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 4 & 4 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 6 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 4 & 4 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 6 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 4 & 4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 6 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 4 & 4 & 0 & 0 \end{pmatrix}.$$

The states are the eleven consecutive univariate double-shell samples; rows alternate the cubic vertex mask $\frac{1}{8}(1, 6, 1)$ and the midpoint mask $\frac{1}{2}(1, 1)$, with boundary zeros as displayed. In particular, every row of B sums to one. Six singleton components with zero diagonal surround the nonzero core on one-based indices 4, $\dots$, 8. That core is

$$\frac{1}{8} \begin{pmatrix} 1 & 6 & 1 & 0 & 0 \\ 0 & 4 & 4 & 0 & 0 \\ 0 & 1 & 6 & 1 & 0 \\ 0 & 0 & 4 & 4 & 0 \\ 0 & 0 & 1 & 6 & 1 \end{pmatrix}.$$

Its first and last columns split off two factors $\lambda - 1/8$; the remaining 3×3 determinant is exactly (29). Consequently,

$$(31) \quad \det(\lambda I - B) = \lambda^6 (\lambda - 1) \left(\lambda - \frac{1}{2} \right) \left(\lambda - \frac{1}{4} \right) \left(\lambda - \frac{1}{8} \right)^2.$$

The zero eigenvalue of B has nullity four, so B is not diagonalizable at zero. Concretely,

$$\text{rank } B = 7, \quad \text{rank } B^2 = \text{rank } B^3 = 5,$$

which yields two size-two and two size-one Jordan blocks at zero. Nothing below requires diagonalizability there: the eigenvalues 1 and $1/2$ of B are simple. Thus the displayed B has nonsemisimple zero-primary structure. We

do not use the English translation's general geometric-multiplicity wording at zero; that translation was machine-produced and explicitly not content-checked. The half-spectrum consequence uses only the simple primary spaces of B at 1 and $1/2$, which the exact factorization establishes directly [4, 5, Thm. 2.58].

Dietz proves the Q8 tensor-product identity for the semi-regular initial family in Theorem 6.35 and Corollary 6.36. Hence, up to a simultaneous permutation,

$$(32) \quad V_n = A \otimes S_n.$$

[4, 5, Thm. 6.35 and Cor. 6.36]. For the double-shell family, Theorem 6.38 states the analogous Q8 identity with a proof sketch. We therefore record its spectral consequence separately rather than claiming an entrywise reconstruction of the shell.

Corollary 9.1 (Semi-regular initial-volume half-spectrum). *For every $n \geq 3$, V_n has a simple dominant eigenvalue 1, a semisimple subdominant eigenvalue $1/2$ of algebraic and geometric multiplicity exactly three, and every other eigenvalue has modulus strictly smaller than $1/2$.*

Proof. The matrix A is diagonalizable with eigenvalues $1, 1/2, 1/4$, so the assertion follows by decomposing $A \otimes S_n$ as

$$S_n \oplus \frac{1}{2}S_n \oplus \frac{1}{4}S_n.$$

□

Corollary 9.2 (Conditional double-shell half-spectrum). *Assume the Q8 identity asserted in [4, 5, Thm. 6.38], namely that $\widehat{V}_n$ is permutation-similar to $B \otimes \widehat{S}_n$. Then $\widehat{V}_n$ has a simple dominant eigenvalue 1, a semisimple subdominant eigenvalue $1/2$ of algebraic and geometric multiplicity exactly three, and every other eigenvalue has modulus strictly smaller than $1/2$.*

Proof. Under the stated assumption, (31) and Corollary 7.3 show that a product eigenvalue can equal $1/2$ only in the two cases

$$1 \cdot \frac{1}{2} \quad (\text{twice}), \quad \frac{1}{2} \cdot 1 \quad (\text{once}).$$

In Jordan form, every block participating in these products has size one: 1 and $1/2$ are simple for B , 1 is simple for $\widehat{S}_n$, and $1/2$ is semisimple for $\widehat{S}_n$. Equivalently, applying the commuting Jordan–Chevalley decompositions to $B \otimes I$ and $I \otimes \widehat{S}_n$ shows that a tensor product of primary blocks contributes at eigenvalue $1/2$ only when the product of their semisimple eigenvalues is $1/2$; in the three cases above both nilpotent parts vanish. Hence the $1/2$ -primary part of $B \otimes \widehat{S}_n$ is semisimple and three-dimensional. The possible nontrivial Jordan structure of B at zero can contribute only the product eigenvalue zero. All other products have modulus below $1/2$, and the only product of modulus one is the product of the two simple eigenvalues 1. □

10. CERTIFICATION AND FORMALIZATION

The accompanying artifact contains four complementary layers of checking.

- The primary verifier, `audit_cubic.py`, imports no provenance-only development module. When pointed at the pinned upstream files, which are not redistributed, it authenticates all three complete files and checks selected decisive source fragments. In every mode it checks exact scalar identities, verifies the displayed constant-mode formulas as matrix exponentials, reconstructs the initial matrix by two routes, audits the phase-free outer core, proves the ranks of B, B^2, B^3 , and checks the spectral consequence of the conditional $B \otimes \hat{S}_n$ identity at $n = 3, 4, 5, 7$. Its finite floating-point sweeps are implementation QA only.
- Hardened corroborating copies of the supplied scripts provide an independently organized symbolic expansion, source-style dense reconstruction, and released one-based double-ring reconstruction. They are `artifact/corroborating/symbolic_certificate.py`, `artifact/corroborating/verify_surface_spectrum.py`, and `artifact/corroborating/verify_double_ring_spectrum.py`. The surface verifier checks both coordinate residuals in the projected local diagonal step. The double-ring verifier uses the eleven-state factor B for its conditional product-spectrum check. Explicit runtime checks remain active under optimized Python, and generated certificates are retained for comparison.
- The Lean 4.30 project builds against a fixed mathlib revision [10, 17]. It kernel-checks the nonzero-mode characteristic identity, tangent and Jury-margin factorizations, constant-mode factorization, exclusion of roots outside the scaled interval $(-1, 1)$, the phase-free real 3×3 outer-core factorization, and the characteristic polynomial of A . Its statement about B is only a polynomial reassociation using an externally supplied factorization; it does not compute the determinant of the explicit 11×11 matrix. That determinant and the zero-primary ranks are checked by the exact SymPy verifier. A complex-phase identity is supplemental.
- The manuscript supplies the remaining mathematical interfaces: source-to-matrix translation, cyclic direct-sum accounting, the trigonometric strictness and small-valence cases, the outer support decomposition, and the tensor-product Jordan argument. These interfaces are stated explicitly rather than presented as part of the Lean coverage.

The numerical verifiers are implementation cross-checks, not evidence on which the theorem depends. The surface proof rests on the displayed exact identities and block decompositions. The conditional shell-product check starts from the eleven-state factor B in (30); it does not construct the actual volumetric shell or prove the Q8 tensor identity. The outer surface

argument is real and phase-free; the complex Fourier identity is only redundant verification.

11. SCOPE BOUNDARY: FULLY IRREGULAR VOLUMES

Dietz's fully irregular volumetric matrices have four layers (P, E, F, o) and geometry-dependent lower and upper blocks. They still satisfy the generator identities that produce three independent $1/2$ -eigenvectors, but arbitrary polyhedral combinatorics destroy the cyclic Fourier decomposition used above. The present proof therefore does *not* establish subdominance or exact multiplicity for that family. It settles:

- (1) every generalized cubic surface initial-element matrix S_n of arbitrary valence $n \geq 3$;
- (2) every released generalized cubic surface double-ring matrix $\hat{S}_n$;
- (3) the semi-regular initial-volume tensor product V_n ; and
- (4) , conditional on Dietz's Q8 identity, the double-shell product $\hat{V}_n$.

A proof for fully irregular volumes would require a new invariant decomposition or a uniform contraction theorem for the geometry-dependent four-layer up/down product. Treating that claim as solved would exceed what the present argument proves.

There is also a separate geometric boundary. The spectral conclusions proved here are the missing eigenvalue conditions used in subdivision analysis; by themselves they do not prove regularity and injectivity of a characteristic map, nor a complete C^1 -smoothness theorem for every generated surface. Those geometric properties require their own hypotheses and arguments.

12. CONCLUSION

The empirical half-spectrum claim for Dietz's generalized cubic surface matrices has an exact explanation. For the initial element, the matrix-exponential product becomes one cubic factorization and a family of quadratic inequalities after cyclic diagonalization. For the released double ring, the outer regular stencils reduce to a phase-independent 3×3 core and eigenvalues of modulus at most $1/8$. In both surface categories the two coordinate modes are the only modes that attain $1/2$, their eigenvalues are semisimple, and every remaining eigenvalue is explicitly separated. The proved initial tensor identity yields the expected triple half-spectrum for V_n ; the same conclusion for the double shell is conditional on the Q8 identity.

The formula also identifies why finite experiments observed eigenvalues extremely close to $1/2$: the $k = 2$ mode approaches it at rate $3\pi^2/(4n^2)$. The unresolved part of the original cubic program is now sharply localized to fully irregular three-dimensional combinatorics and, separately, to any geometric characteristic-map conclusions not already supplied by other arguments.

Reproducibility statement. All algebraic identities central to the surface proof are duplicated in exact symbolic form, and independently organized verifiers reconstruct the initial and double-ring surface matrices. The Lean source contains no `sorry`, `admit`, or project-declared axiom. The artifact records the precise scope of the formalization, its standard foundational dependencies, and the commands required for kernel checking. The versioned preprint, source, exact certificates, verification software, and Lean companion are available in the Zenodo record at [doi:10.5281/zenodo.21925578](https://doi.org/10.5281/zenodo.21925578).

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